Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



MerchReflex

Team: Finfold Growth Lab

Team leader: Yue Zhao

Retail & Commerce: evidence-backed page maintenance

Implementation draft · 13 October 2026



AI-generated image · synthetic product

Fix the page.
Not just the ticket.

Customer feedback becomes a precise page repair, with visible evidence and a way back.

Brief about the idea

A merchant receives the same product question again.

MerchReflex connects that question to the page, the product image and merchant-confirmed facts.

It proposes the smallest supported change, requests missing facts, and verifies the published version.

Opportunities

Existing assistants already edit products and request approval.

Our focus: distinguish page confusion from quality or delivery problems; resume after missing facts arrive; expire outdated approvals; verify the served result.

The hypothesis is less manual page-maintenance work. Merchant value remains to be measured.

Features in the working prototype

One product. Three editable fields.

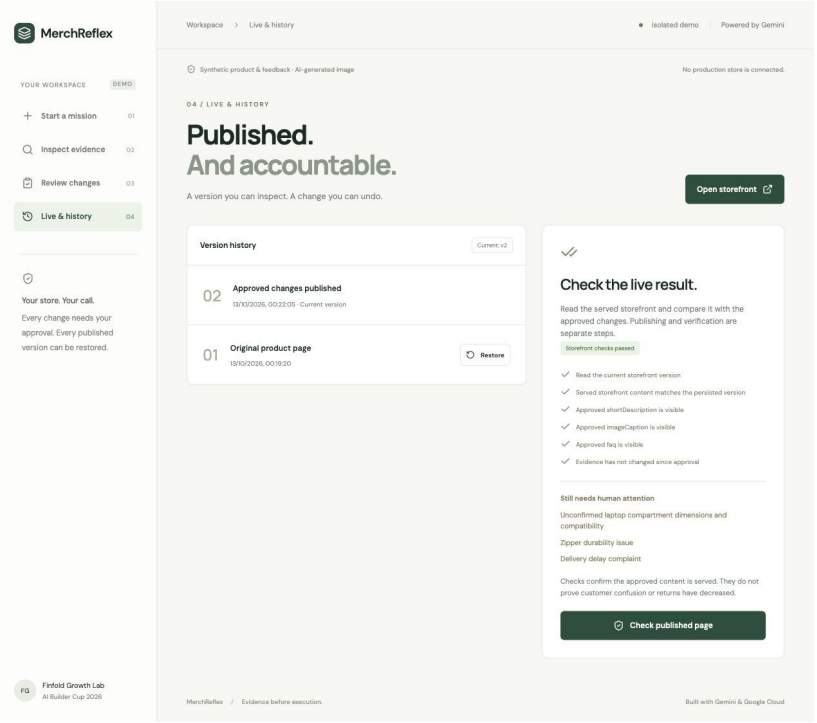
Source-linked classification and minimal field differences
Merchant-confirmed supplements and reassessment
Approval bound to evidence, content and page versions
Transactional publication, served-page checks and rollback

Price, inventory, policy and certification edits are blocked.

Repair workflow

01	**Evidence**	Feedback + specification + image
02	**Judgment**	Repair / request facts / conflict / human action
03	**Review**	Minimal patch + a separate semantic check
04	**Approval**	Merchant accepts exact text and source revision
05	**Execution**	Publish → read served page → verify or restore

Verification interface

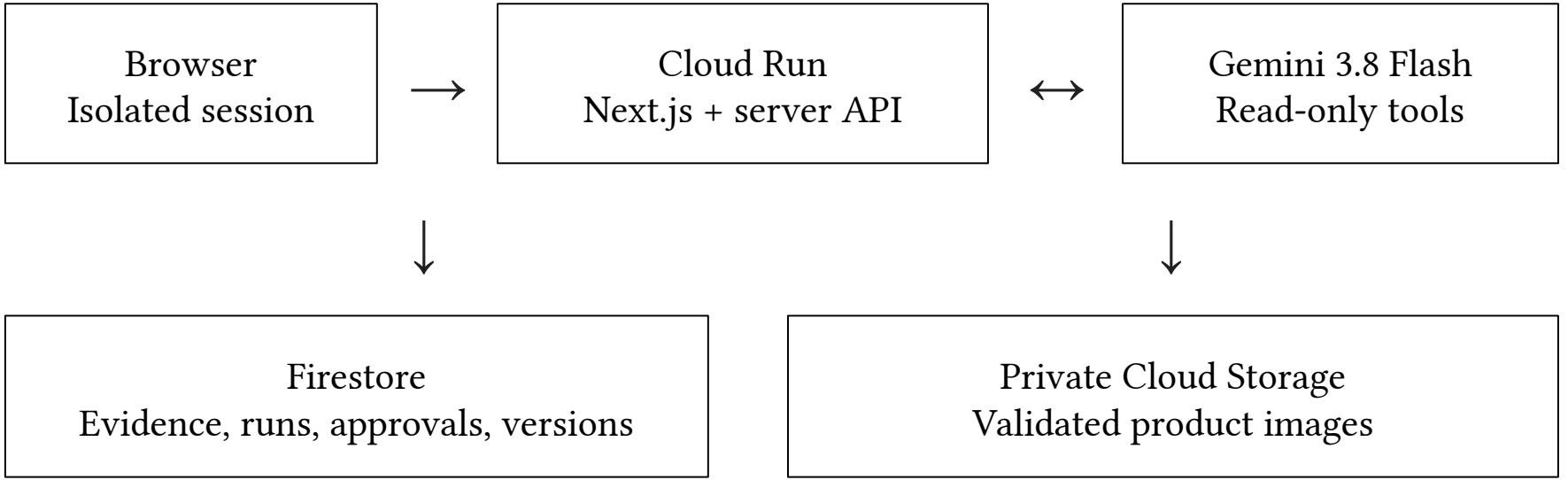


Published content is read back from the storefront.

Checks compare the actual served version with the approved changes.

Unresolved issues remain visible.

Google Cloud architecture



Cloud Run handles writes and verification. The model has no publish tool.

Technologies used

Gemini 3.8 Flash through Google Gen AI SDK
Text, image, function calling and structured output

Next.js + TypeScript on Cloud Run
Firestore transactions + private Cloud Storage

Strict schemas, server-side limits, exact-origin checks, CSRF protection and HttpOnly session cookies

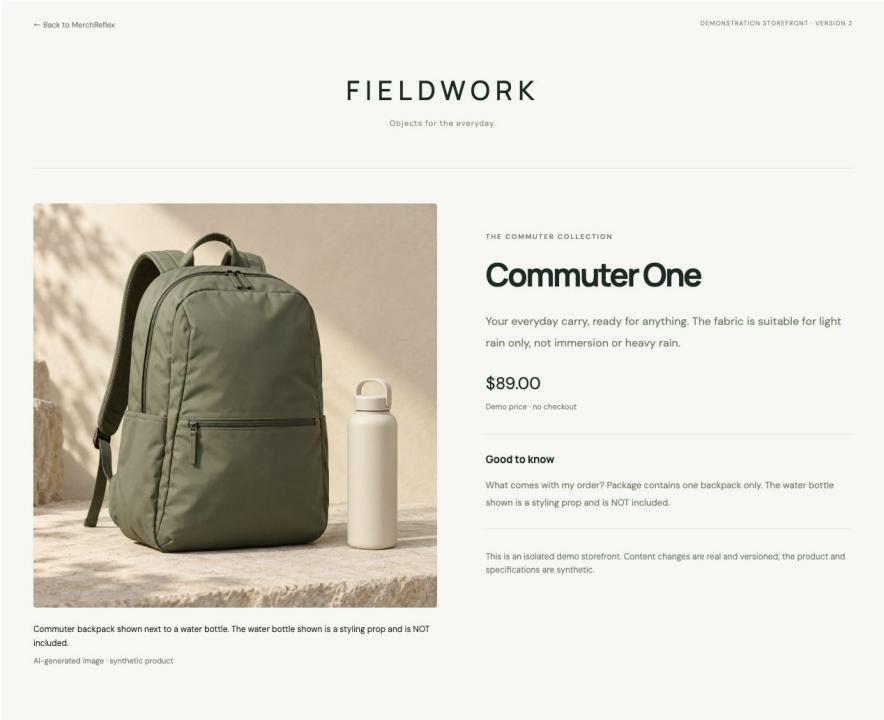
Prototype operating budget

CNY 500 total planning allowance

CNY 200 Model development and evaluation
CNY 150 Deployment, storage and judging availability
CNY 150 Reserve

Global model quota, daily limits, maximum two instances.
Budget alerts are not hard billing caps. No credits assumed.

Published prototype



Observed on the live storefront:

The unsupported waterproof promise was replaced.

The image caption and FAQ clarify that the bottle is excluded.

Synthetic demo; no checkout.

Performance and evaluation

23 execution and integrity tests pass locally.
Live publication, served-page verification and restoration have been exercised in the browser.

30 semantic cases: 18 development, 12 frozen holdout.
Same-model, single-response baseline.

Human semantic grading and merchant task testing remain pending. No accuracy or revenue claim is made.

Future development

Before submission
Complete blind human grading, two task tests and merchant interviews; disclose failures.

After qualification
Use authorized merchant data, connect one development store, then test ongoing page maintenance.

Measure adoption and time saved before pricing or ROI claims.

Public prototype and source

GitHub
https://github.com/joeymilano/merchreflex

Live product
https://merchreflex-927 668 522 599.asia-southeast1.run.app

English demonstration video
Recording and final public link pending

What the evidence does and does not show

Synthetic product facts are labelled as merchant-confirmed demo data, not certifications.

Published content can be read, checked and restored.
A passed check does not prove lower returns or higher revenue.

Same-model verification can share mistakes.
The merchant retains the final decision.

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Thank you

